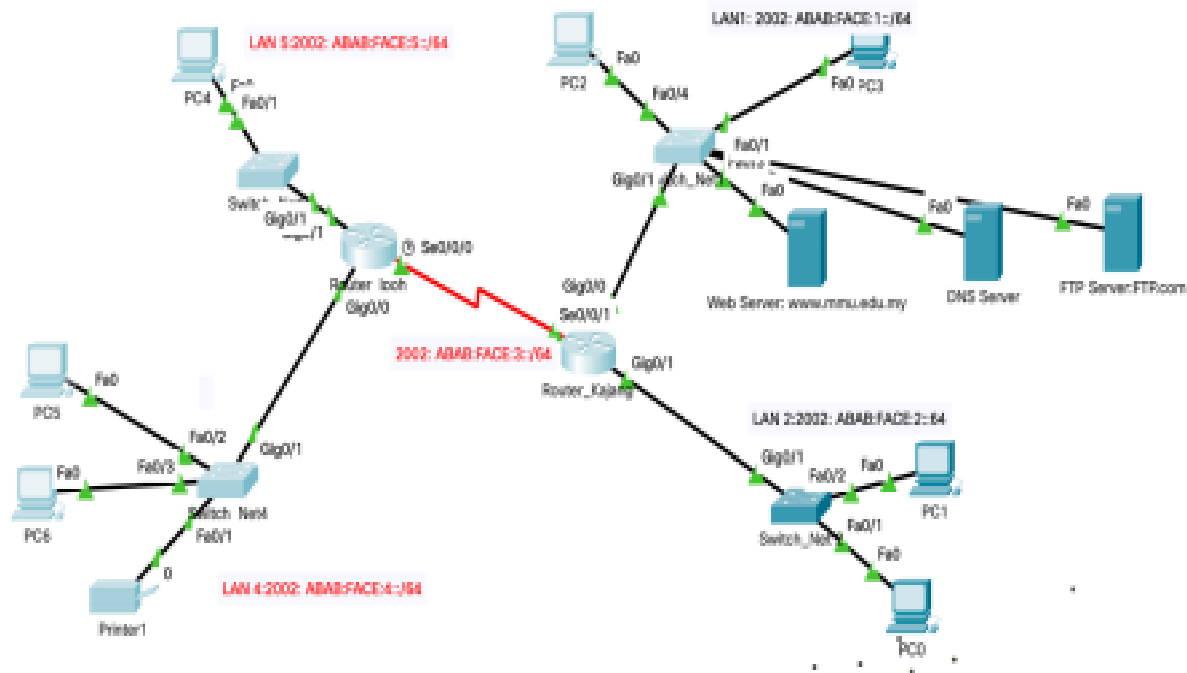


TCN2141 Computer Networks

Lab 5 (6 marks):

Working with the Application Layer: DHCPv6, DNS, FTP and HTTP

Important instruction: Print lab sheet and submit it with all solution derived from lab experiment to tutors.



Addressing Table

Device	Interface	IPv6 Address	Link-Local
Router Kajang	G0/0	2002: ABAB:FACE:1::1/64	FE80::1:1
	G0/1	2002: ABAB:FACE:2::1/64	FE80::2:1
	S0/0/1	2002: ABAB:FACE:3::1/64	FE80::3:1
Router Ipoh	G0/0	2002: ABAB:FACE:4::1/64	FE80::4:1
	G0/1	2002: ABAB:FACE:5::1/64	FE80::5:1
	S0/0/0	2002: ABAB:FACE:3::2/64	FE80::3:2

Device	Interface	IPv6 Address	Default Gateway
PC1	NIC	DHCPv6 Assigned	DHCPv6 Assigned
PC2	NIC	DHCPv6 Assigned	DHCPv6 Assigned
PC3	NIC	DHCPv6 Assigned	DHCPv6 Assigned
PC4	NIC	DHCPv6 Assigned	DHCPv6 Assigned

PC5	NIC	DHCPv6 Assigned	DHCPv6 Assigned
PC6	NIC	DHCPv6 Assigned	DHCPv6 Assigned
Printer1	NIC	2002: ABAB:FACE:4::100/64	2002: ABAB:FACE:4::1
DNS Server	NIC	2002: ABAB:FACE:1::200/64	2002:ABAB:FACE:1::1
Web Server	NIC	2002: ABAB:FACE:1::3/64	2002:ABAB:FACE:1::1
FTP Server	NIC	2002: ABAB:FACE:1::4/64	2002:ABAB:FACE:1::1

Objectives

Part 1: Configure a Router

- a) Basic configuration
- b) DHCPv6 server -Pool
- c) Configure interfaces with correct ipv6 address
- d) Configure DHCPv6 stateful
- e) Configure OSPF routing

Part 2: Configure DNS server

Part 3: Configure Web Server

Part 3: Configure FTP Server

Part 4: Verify DHCP , DNS, FTP and HTTP Connectivity

Instructions:

Task 1

1. Start Packet Tracer using **Real-time** mode.
 - Options -> Preferences
 - Enable "Show Link Lights"
 - Disable "Hide Device Label"
 - Enable 'Always show port label'

Task 2

Configure **Router**

Router Kajang

1. Access global configuration mode
Router > Enable
Router # Configure terminal
2. Change router name
Router (config)#hostname Router_Kajang
3. Activate IPv6 functions
Router_Kajang(config)# ipv6 unicast-routing
Router_Kajang(config)#
4. Create DHCPv6 stateful Pool
Router_Kajang(config)# ipv6 dhcp pool **STATEFUL_1**
Router_Kajang(config-dhcpv6)#address prefix 2002:ABAB:FACE:1::/64
Router_Kajang(config-dhcpv6)#dns-server 2002:ABAB:FACE:1::200
Router_Kajang(config-dhcpv6)# domain-name FIST_Lab.com
Router_Kajang(config-dhcpv6)#exit

Router_Kajang(config-dhcpv6)#ipv6 dhcp pool STATEFUL_2
Router_Kajang(config-dhcpv6)#address prefix 2002:ABAB:FACE:2::/64
Router_Kajang(config-dhcpv6)#dns-server 2002:ABAB:FACE:1::200
Router_Kajang(config-dhcpv6)#domain-name FIST_Lab.com
Router_Kajang(config-dhcpv6)#exit
Router_Kajang(config-dhcpv6)#
5. Configure OSPF Routing
Router_Kajang(config-rtr)#ipv6 router ospf 10
Router_Kajang(config-rtr)#router-id 1.1.1.1
Router_Kajang(config-rtr)#exit
Router_Kajang(config)#

6. Interfaces with IPv6 address and Link-local address. The ipv6 nd managed-config flag sets a flag in the router advertisement that tells the hosts that they could use DHCPv6. To enable the OSPF-for-IPv6 process on an interface and assign that interface to an area, use the **ipv6 ospf ospf-process area area-id** command

```
Router_Kajang(config-if)#interface GigabitEthernet0/0
Router_Kajang(config-if)#ipv6 address
2002:ABAB:FACE:1::1/64
Router_Kajang(config-if)#ipv6 address FE80::1:1 link-local
Router_Kajang(config-if)# ipv6 nd managed-config-flag
Router_Kajang(config-if)# ipv6 dhcp server STATEFUL_1
Router_Kajang(config-if)#ipv6 ospf 10 area 0
Router_Kajang(config-if)#no shut
Router_Kajang(config-if)#exit
```

```
Router_Kajang(config-if)#interface GigabitEthernet0/1
Router_Kajang(config-if)#ipv6 address FE80::2:1 link-local
Router_Kajang(config-if)#ipv6 address
2002:ABAB:FACE:2::1/64
Router_Kajang(config-if)# ipv6 nd managed-config-flag
Router_Kajang(config-if)#ipv6 ospf 10 area 0
Router_Kajang(config-if)#ipv6 dhcp server STATEFUL_2
Router_Kajang(config-if)#no shut
Router_Kajang(config-if)#exit
```

```
Router_Kajang(config-if)#interface Serial0/0/1
Router_Kajang(config-if)# ipv6 address
2002:ABAB:FACE:3::1/64
Router_Kajang(config-if)#ipv6 address FE80::3:1 link-local
Router_Kajang(config-if)# ipv6 enable
Router_Kajang(config-if)#ipv6 ospf 10 area 0
Router_Kajang(config-if)#no shut
Router_Kajang(config-if)#end
```

7. Save Configuration to NVRAM

```
Router_Kajang#copy run start
Destination filename [startup-config]? Enter
Router_Kajang#
```

Router Ipoh

1. Access global configuration mode

```
Router > Enable
Router # Configure terminal
```

2. Change router name

```
Router (config)#hostname Router_Ipoh
```

3. Activate IPv6 functions

```
Router_Ipoh (config)# ipv6 unicast-routing
Router_Ipoh (config)#
```

4. Create DHCPv6 stateful Pool

```
Router_Ipoh (config)# ipv6 dhcp pool STATEFUL_4
Router_Ipoh (config-dhcpv6)#address prefix 2002:ABAB:FACE:4::/64
Router_Ipoh (config-dhcpv6)#dns-server 2002:ABAB:FACE:1::200
Router_Ipoh (config-dhcpv6)# domain-name FIST_Lab.com
Router_Ipoh ajang(config-dhcpv6)#exit
```

```
Router_Ipoh (config-dhcpv6)#ipv6 dhcp pool STATEFUL_5
Router_Ipoh (config-dhcpv6)#address prefix 2002:ABAB:FACE:5::/64
Router_Ipoh (config-dhcpv6)#dns-server 2002:ABAB:FACE:1::200
Router_Ipoh (config-dhcpv6)#domain-name FIST_Lab.com
Router_Ipoh (config-dhcpv6)#exit
Router_Ipoh (config-dhcpv6)#
```

5. Configure OSPF Routing

```
Router_Ipoh (config-rtr)#ipv6 router ospf 10
Router_Ipoh (config-rtr)#router-id 2.2.2.2
Router_Ipoh (config-rtr)#exit
Router_Ipoh (config)#
```

6. Interfaces with IPv6 address and Link-local address. The ipv6 nd managed-config flag sets a flag in the router advertisement that tells the hosts that they could use DHCPv6. To enable the OSPF-for-IPv6 process on an interface and assign that interface to an area, use the **ipv6 ospf ospf-process area area-id** command

```
Router_Ipoh (config-if)#interface GigabitEthernet0/0
Router_Ipoh (config-if)#ipv6 address 2002:ABAB:FACE:4::1/64
Router_Ipoh (config-if)#ipv6 address FE80::4:1 link-local
Router_Ipoh (config-if)# ipv6 nd managed-config-flag
Router_Ipoh (config-if)# ipv6 dhcp server STATEFUL_4
Router_Ipoh (config-if)#ipv6 ospf 10 area 0
Router_Ipoh (config-if)#no shut
```

```
Router_Ipoh (config-if)#exit
```

```
Router_Ipoh (config-if)#interface GigabitEthernet0/1
```

```
Router_Ipoh (config-if)#ipv6 address
```

```
2002:ABAB:FACE:5::1/64
```

```
Router_Ipoh (config-if)#ipv6 address FE80::5:1 link-local
```

```
Router_Ipoh (config-if)# ipv6 nd managed-config-flag
```

```
Router_Ipoh (config-if)#ipv6 ospf 10 area 0
```

```
Router_Ipoh (config-if)#ipv6 dhcp server STATEFUL_5
```

```
Router_Ipoh (config-if)#no shut
```

```
Router_Ipoh (config-if)#exit
```

```
Router_Ipoh (config-if)#interface Serial0/0/0
```

```
Router_Ipoh (config-if)# ipv6 address
```

```
2002:ABAB:FACE:3::2/64
```

```
Router_Ipoh (config-if)#ipv6 address FE80::3:2 link-local
```

```
Router_Ipoh (config-if)# ipv6 enable
```

```
Router_Ipoh (config-if)#ipv6 ospf 10 area 0
```

```
Router_Ipoh (config-if)#no shut
```

```
Router_Ipoh (config-if)#end
```

7. Save Configuration to NVRAM

```
Router_Ipoh #copy run start
```

```
Destination filename [startup-config]? Enter
```

```
Router_Ipoh #
```

Task 3

Configuring the DNS Server

a) Add a server.

Global Settings:

- Change the Display Name to “DNS Server”
- Set the Gateway to **2002: ABAB: FACE:1::1/64**
- Set Link Local: **FE80::1:200**

b) Fast Ethernet:

- Set the IP address to **2002: ABAB: FACE:1::200/64**
-

c) HTTP:

- Set HTTP Service and HTTPS Service to **Off**

d) DHCP:

- Set the Service to **Off**

e) DNS:

- Entering the **www.mmu.edu.my** Domain Name
 - Enter for the Domain Name **www.mmu.edu.my**
 - Enter for IP Address **2002: ABAB: FACE:1::4/64**
 - Click **Add**
- Entering FTP server Domain Name

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- Enter for the Domain Name **FTP.com**
- Enter for IP Address **2002: ABAB: FACE:1::5/64**
- Click **Add**

Task 4

Configuring the Web Server: **www.mmu.edu.my**

a) Add a server.

Global Settings:

- Change the Display Name to “**Web Server: www.mmu.edu.my**”
- Set the Gateway to **2002: ABAB: FACE:1::1/64**

b) Fast Ethernet:

- Set the IP address to **2002: ABAB: FACE:1:4/64**
-

c) DHCP:

- Set the Service to **Off**

d) DNS:

- Set the Service to **Off**

e) HTTP

Change the sentence,

<html>

<center>**Multimedia University**</center>

<hr>Welcome to FIST: Computer Network. Opening doors to new opportunities.

Mind Wide Open.

</html>

You may add other information as well.

Task 5

Configuring the FTP Server: **FTP.com**

a) Add a server.

Global Settings:

- Change the Display Name to “**FTP Server: FTP.com**”
- Set the Gateway to **2002: ABAB: FACE:1::1/64**

b) Fast Ethernet:

- Set the IP address to **2002: ABAB: FACE:1::5/64**
-

c) DHCP:

- Set the Service to **Off**

d) DNS:

- Set the Service to **Off**

e) FTP

- Set username: **admin**
- Set password: Lab5
- Select -check Read Write Delete Rename and List

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- Click **Add**

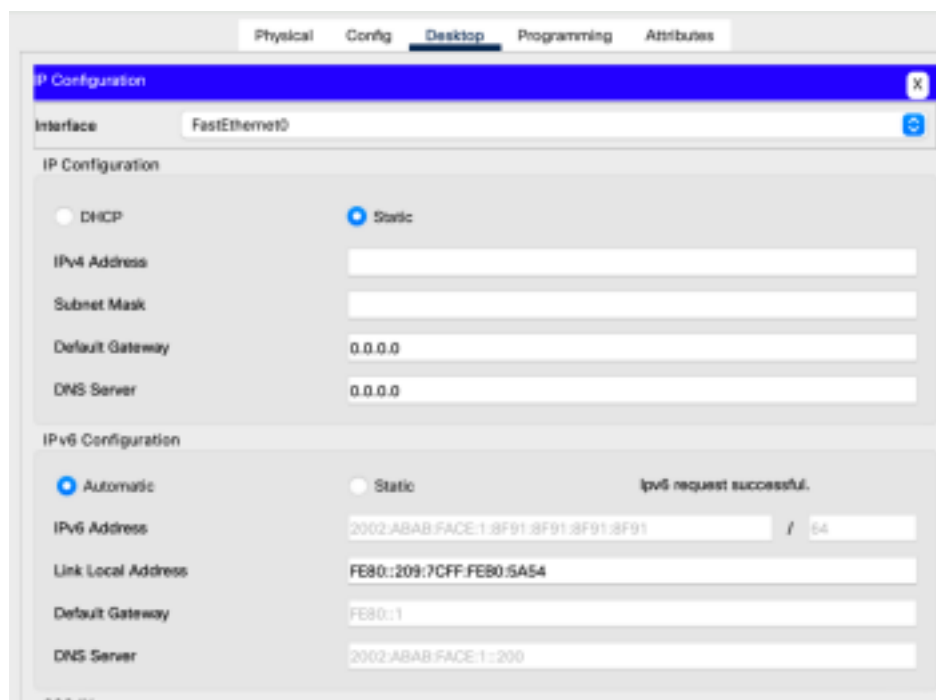
Task 6

Configure all Client Computers using DHCP in all LAN in Router Kajang and Router Ipoh

a) Click PC

b) Go to Desktop > IP Configuration

c) Select IPv6 Configuration > click automatic button



Task 7

Configure static IPv6 address for Printer in Lan 4

a) Set Fast Ethernet - IPv6 address : **2002:ABAB:FACE:4::100**

b) Set Link-local address : **FE80::4:100**

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c) Set Default Gateway: **2002:ABAB:FACE:4::1**

d) Set DNS: **2002:ABAB:FACE:1::200**

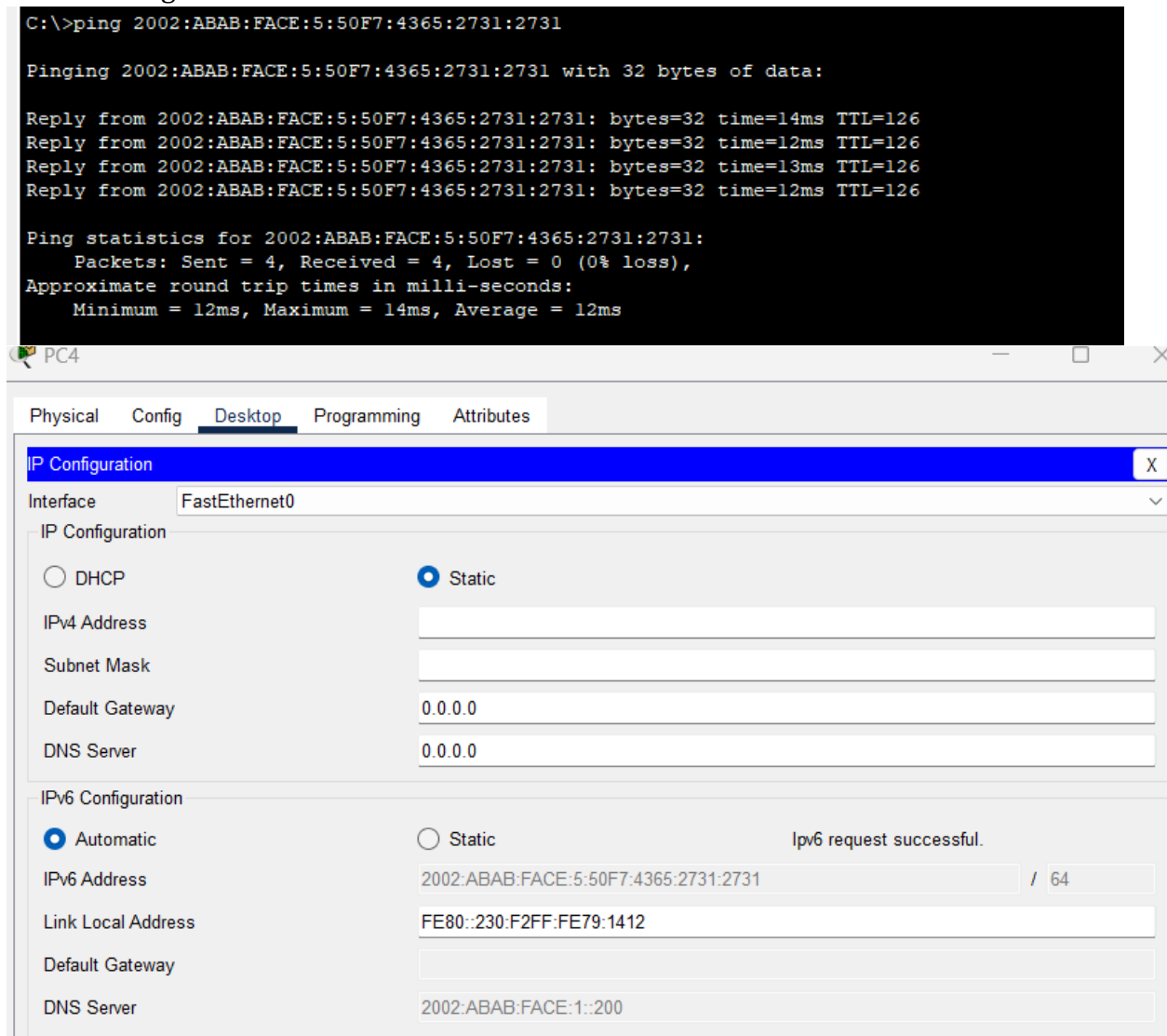
Task 8

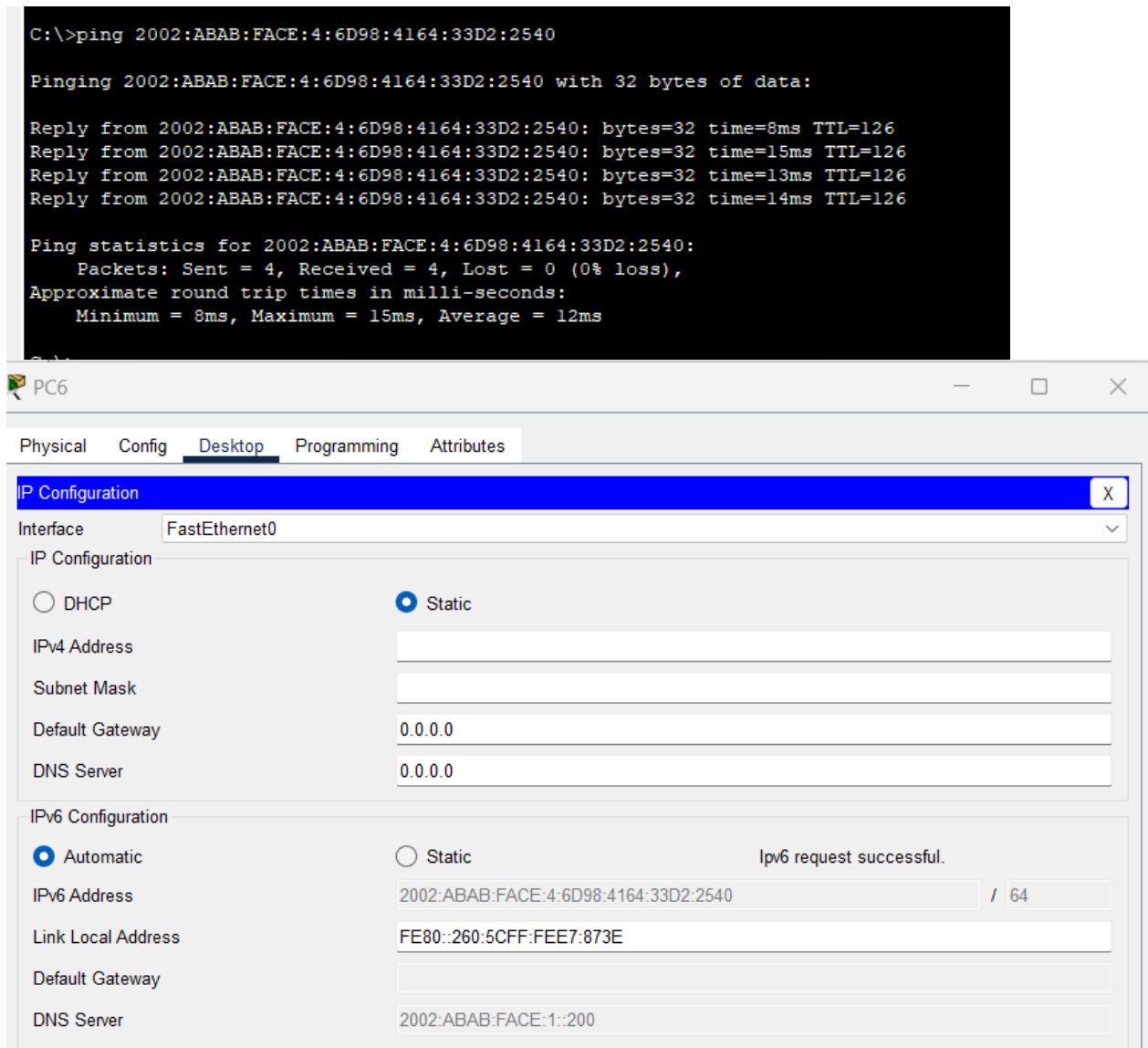
Verify connectivity

Add Screenshot to proof your activities

a) Ping (ICMP) -

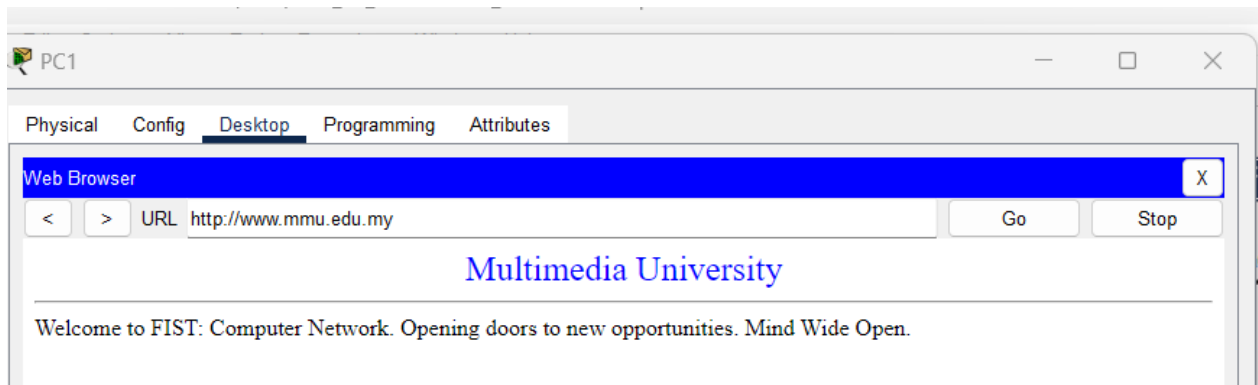
- From a client computer use the Desktop Command prompt to ping the other client computers and the servers.
- Example: From the PC1 client, C> **ping PC 4, PC6**
- The first one or two pings may fail, but you should receive a reply on the later pings. This is due to the ping timing out while the ARP process takes place (later).
- Show Ping results





b) Web Browser (HTTP)

- On the client computers use the Desktop Web Browser, enter the URLs of the Web Servers www.mmu.edu.my a
- You should see the web pages that you created on these server.
- Show Browser results



c) FTP Server

- Login from PC1
- Create a text file using text editor in desktop
- Upload a txt file to ftp server : ftp> put Filename.txt
- Make sure the file is in FTP server

FTP Server:FTP.com

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP**
- IoT
- VM Management
- Radius EAP

FTP

Service ☒ On ☐ Off

User Setup

Username Password

☒ Write ☒ Read ☒ Delete ☒ Rename ☒ List

	Username	Password	Permission
1	admin	Lab5	RWDNL
2	cisco	cisco	RWDNL

Add Save Remove

File

28	ir800-universalk9-mz.SPA.155-3.M
29	ir800-universalk9-mz.SPA.156-3.M
30	ir800_yocto-1.7.2.tar
31	ir800_yocto-1.7.2_python-2.7.3.tar
32	pt1000-i-mz.122-28.bin
33	pt3000-i6q412-mz.121-22.EA4.bin
34	test.txt

Remove

```
Minimum = 0ms, Maximum = 10ms, Average = 12ms

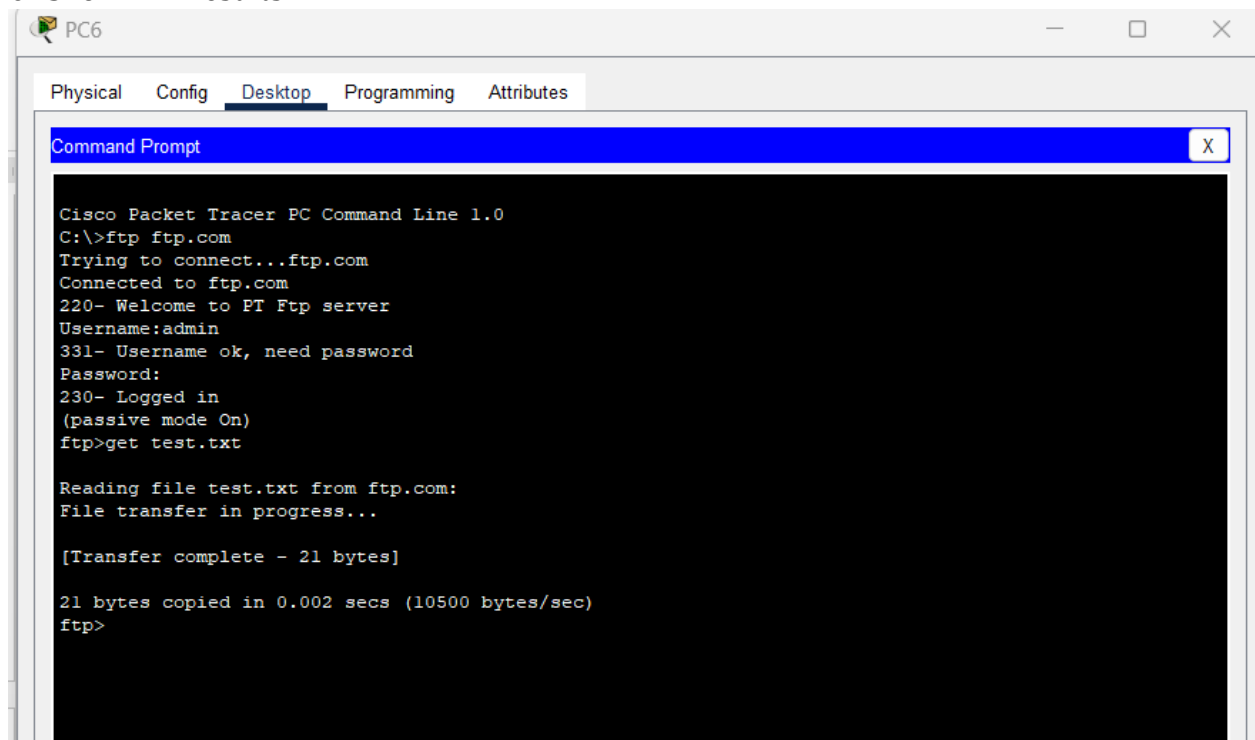
C:\>ftp ftp.com
Trying to connect...ftp.com
Connected to ftp.com
220- Welcome to PT Ftp server
Username:admin
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>put test.txt

Writing file test.txt to ftp.com:
File transfer in progress...

[Transfer complete - 21 bytes]

21 bytes copied in 0.08 secs (262 bytes/sec)
ftp>
```

- Login to FTP server from PC 6
- Download the text file for server: ftp>get Filename.txt
- Open it and display
- Show FTP results



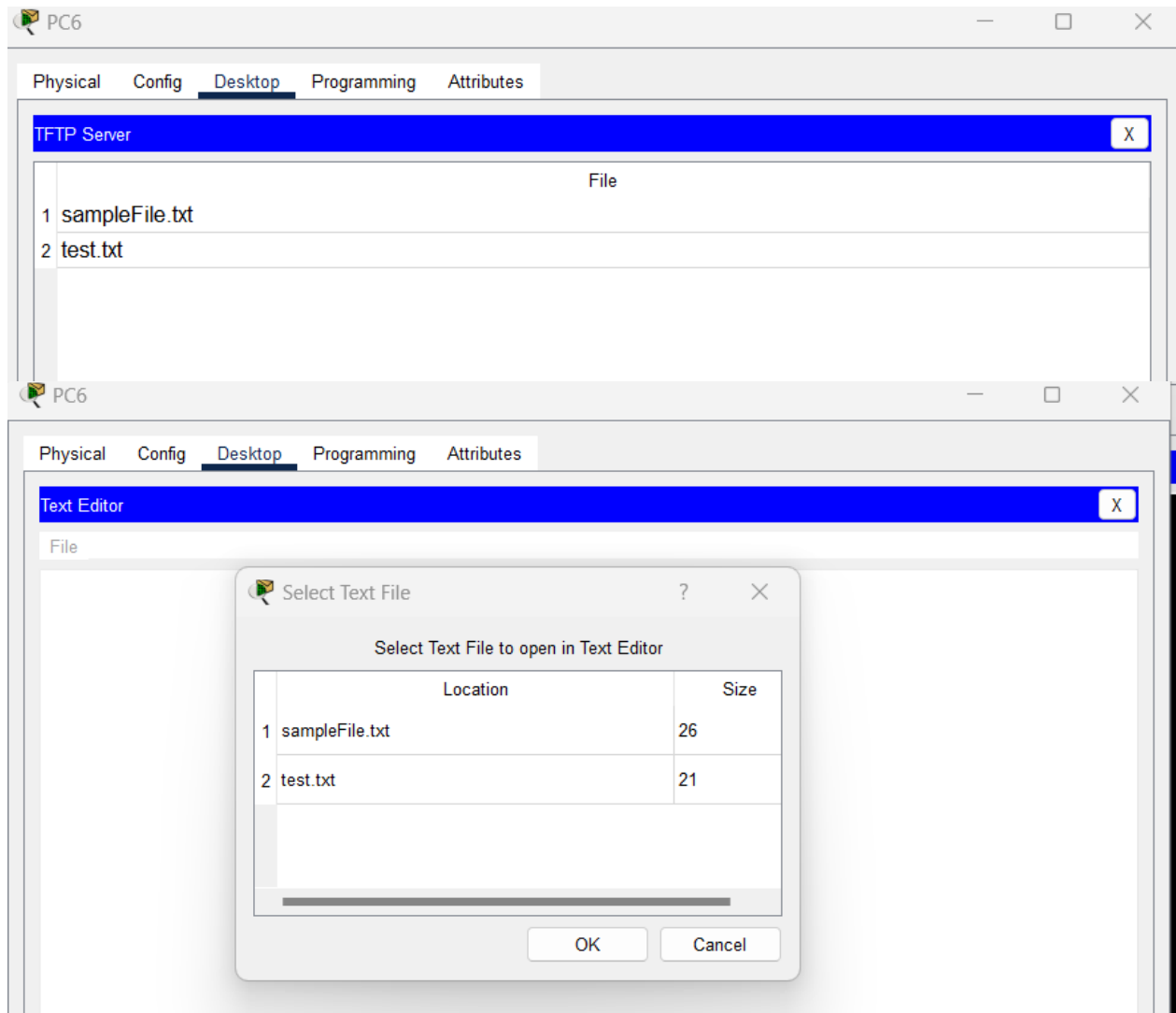
```
PC6
Physical Config Desktop Programming Attributes
Command Prompt X

Cisco Packet Tracer PC Command Line 1.0
C:\>ftp ftp.com
Trying to connect...ftp.com
Connected to ftp.com
220- Welcome to PT Ftp server
Username:admin
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>get test.txt

Reading file test.txt from ftp.com:
File transfer in progress...

[Transfer complete - 21 bytes]

21 bytes copied in 0.002 secs (10500 bytes/sec)
ftp>
```



Task 9

1. Submit Packet Tracer (.pkt) and this Lab 5 Quest with appropriate answers